

İREM ÖZDEMİR

Software Engineer , Python & Backend Development , AI & Automation
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PROFESSIONAL SUMMARY

I am a recent software engineering graduate developing backend projects with Python and Java, who integrates REST API design, data processing and artificial intelligence technologies in internship and academic projects. I have carried out applied studies on computer vision, automation tools and good problem solving skills to keep up with dynamic teams.

EDUCATION

University of Turkish Aeronautical Association, B.Sc. in Software Engineering

2021 – 2026

WORK EXPERIENCE

Intern Network & Telecommunications/Nokia,Alcatel

July/2025 – September/2025

- Gained hands-on technical exposure to mobile and 5G network infrastructure, base stations, and signal transmission processes.
- Benchmarked 8+ document-parsing libraries for table and image extraction performance from PDFs; prepared a reproducible comparison report in Python.
- Performed UI test automation using Selenium; designed and executed test scenarios aligned with Microsoft Certified Professional (MCP) standards.
- Built a Python data-processing tool that analyzes CSV files and generates automated HTML reports with statistical summaries of network devices.

Intern Information Technologies/Presidency of the Republic of Turkey

January/2025 – February/2025

- Gained hands-on experience with IT systems and communication technology processes in a public-sector environment.
- Supported maintenance and optimization of digital infrastructure operations, contributing to data organization and system reliability.

Intern Software Development/KEYDATA Information Technologies Systems

August/2024 – September/2024

- Developed individual software projects to strengthen algorithmic thinking and application development skills.
- Built responsive front-end interfaces using HTML, improving the structure and user experience of web applications.

PROJECTS

AI Lifestyle Coach Agent/ Capstone Project

Tech Stack: Python, FastAPI, OpenCV, MediaPipe, SQLite, Tkinter, Ollama (LLM), REST API

- Built an AI-powered fitness and wellness platform combining real-time exercise tracking, health monitoring, and personalized coaching.
- Developed a computer vision engine with MediaPipe and OpenCV that detects 33 body landmarks, counts repetitions, and provides instant exercise form feedback.
- Implemented real-time posture analysis and automatic calorie estimation (MET and Mifflin-St Jeor formulas) for 7 bodyweight exercises.
- Designed a FastAPI backend with 20+ REST API endpoints and a cross-platform Tkinter desktop interface.
- Managed user health data (exercise, sleep, hydration, calories, steps) in SQLite, generating personalized recommendations via a rule-based engine integrated with a local Ollama LLM.

Photobooth Web Application

Tech Stack: Python, React, TypeScript, JavaScript, Vite, Tailwind CSS, WebRTC API, HTML5

- Developed an interactive, responsive web based photobooth application enabling users to capture snapshots, apply real time visual filters, and download personalized photo strips.
- Integrated browser media devices using the WebRTC API for live webcam streaming and utilized HTML5. Canvas for advanced image rendering, manipulation, and custom frame overlays.
- Configured a modern frontend build pipeline with Vite and deployed the application via Netlify with automated CI/CD synchronization.

AI Job Matcher

Tech Stack: Python, FastAPI, Pandas, PyMuPDF, HTML/CSS, REST API, SQLite

- Developed a local, API-independent intelligent job matching and search engine designed to optimize personal recruitment workflows.
- Designed a FastAPI backend architecture integrating dynamic job scraping sources (JSearch and Jooble) with a custom TF-IDF text analytics engine in Python and Pandas to compute precision-based match scores.
- Implemented PyMuPDF (fitz) integration for automated PDF resume parsing, accompanied by intelligent seniority estimation and asymmetric penalty algorithms for accurate role filtering.
- Integrated SQLite database management to persistently handle user profiles and historical match analyses, delivering an end-to-end functional application.

Web Scraper

Tech Stack: Python, Selenium, pandas, openpyxl

- Built a web scraping tool that automatically collects product details (name, brand, rating, price, link) from Amazon; exported data to Excel using pandas and openpyxl.

Library Management System

Tech Stack: Java, MySQL, JSON, XML, MongoDB

- Designed a desktop-based system to manage book borrowing, returns, and inventory; integrated JSON for book metadata and XML export for borrowing records; used MongoDB to manage reviews and ratings.

Object-Oriented Color & Visual Recognition System

Tech Stack: Python, OpenCV

- Developed an image-processing application for real-time color detection and classification using webcam input.

TECHNICAL SKILLS

- Programming Languages: Python (Automation, Web Scraping, Data Processing), Java (Object-Oriented Programming), C (Basic)
- Backend Development: FastAPI, REST API Design & Development
- AI & Computer Vision: OpenCV, MediaPipe, Ollama (Local LLM)
- Web & Markup Technologies: HTML, JSON, XML
- Databases: SQL, MySQL (Triggers, Stored Procedures), MongoDB (NoSQL)
- Automation & Testing: Selenium (UI Test Automation)
- Mobile Development: Flutter, Dart
- Data Processing & Reporting: pandas, openpyxl, CSV Data Handling and Reporting
- Tools & Platforms: Git, GitHub, GitHub Copilot, Claude AI

LANGUAGES

- Turkish (Native)
- English (B2, Upper Intermediate)